Shared Task Overview and Related Work

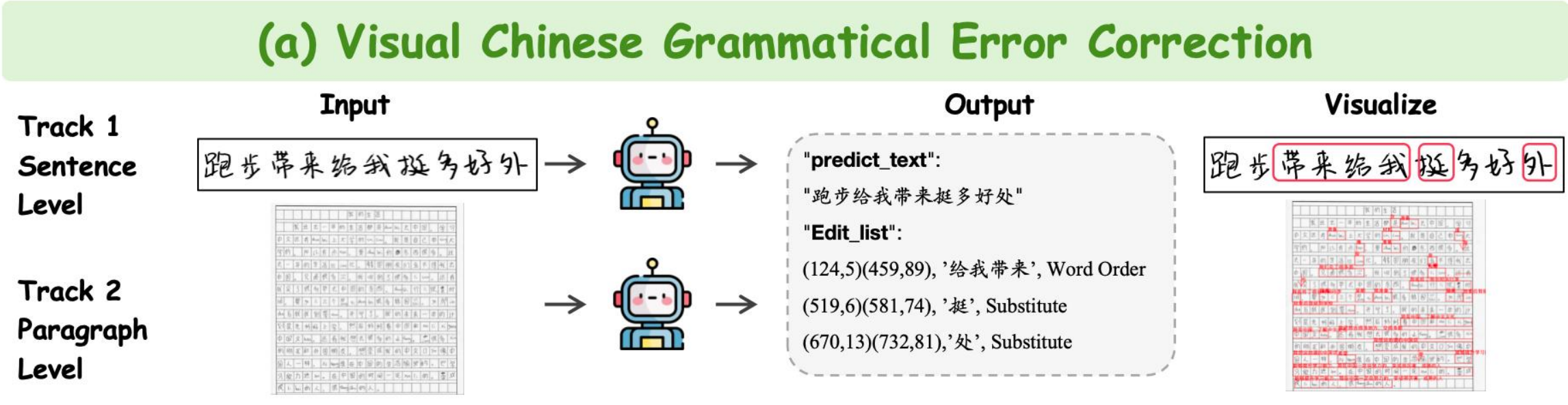
VisCGEC-v2.0

Visual Chinese Grammatical Error Correction

Visual feedback for handwritten Chinese

Visual Chinese grammatical error correction

Handwritten image input



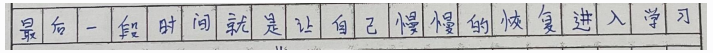
Corrected full text + edits grounded in the original image

Two tracks, one output schema

Sentence-Level Track

Single handwritten sentence

Localized grammatical correction



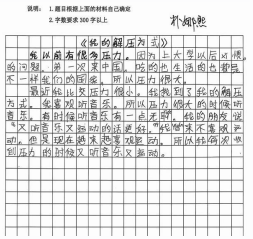


Paragraph-Level Track

Multiple handwritten sentences

Broader discourse context





Images	Train	Dev	Test
Sentence-Level	1,960	245	246
Paragraph-Level	350	—	88

Real handwriting from learners of Chinese as a foreign language

Image input and structured submission

TEST INPUT

img_id + img_path

No source_text at test time

EDIT OPERATIONS

Insert Delete

Substitute WordOrder

DATA AND OUTPUT FORMATS


```
{
 "img_id": "720_8",
 "img_path": "images/720_8.png",
 "target_text": "跑步给我带来挺多好处",
 "edit_list": [
  {
   "bbox": {
    "start_x": 281,
    "start_y": 4,
    "end_x": 345,
    "end_y": 86
   },
   "correct_text": "挺",
   "edit_type": "Substitute"
  },
  ...
 ]
}
```


Data Format

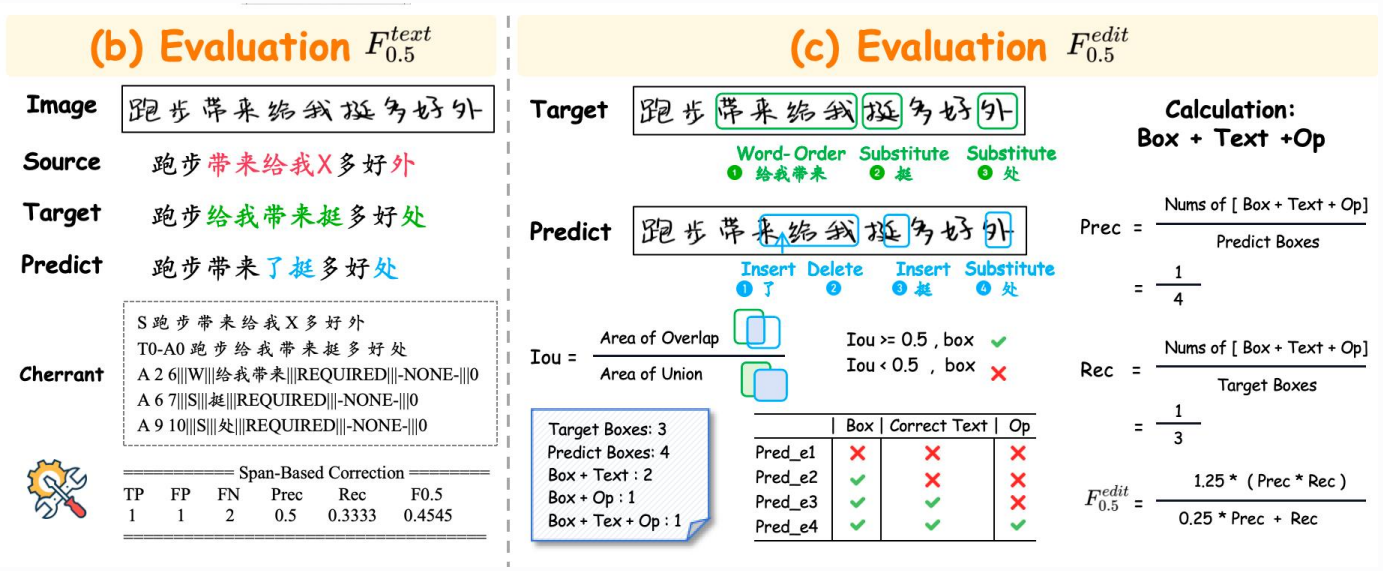

```
[
  {
   "img_id": "xxx",
   "predict_text": "xxx",
   "edit_list": [
    {
     "bbox": {
      "start_x": xxx,
      "start_y": xxx,
      "end_x": xxx,
      "end_y": xxx
     },
     "correct_text": "xxx",
     "edit_type": "xxx"
    },
    ...
   ]
  }
  ...
]
```


Output Format

Evaluation combines text and localized edits

50% text F0.5 + 50% edit F0.5

Compute per sample, then average across samples



All three conditions must hold

Region overlap

Bounding-box IoU ≥ 0.5

Correction text

Exact match with gold correction

Operation type

Match with the gold edit type

Related work

END-TO-END VISCGEC WITH MLLMS

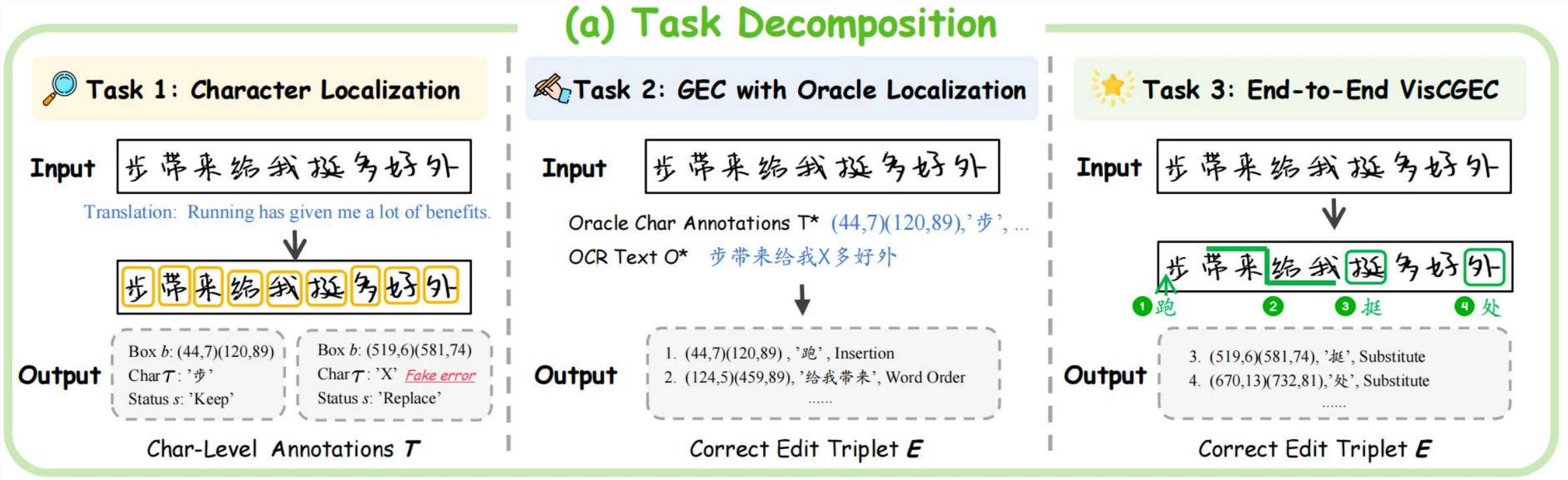
Exploring End-to-End Paradigms for Visual Chinese Grammatical Error Correction with MLLMs



Xiaoman Wang, Wenxuan Huang, Wenbiao Tao, Yike Zhao, Yaohui Liu, Yunshi Lan, Weining Qian

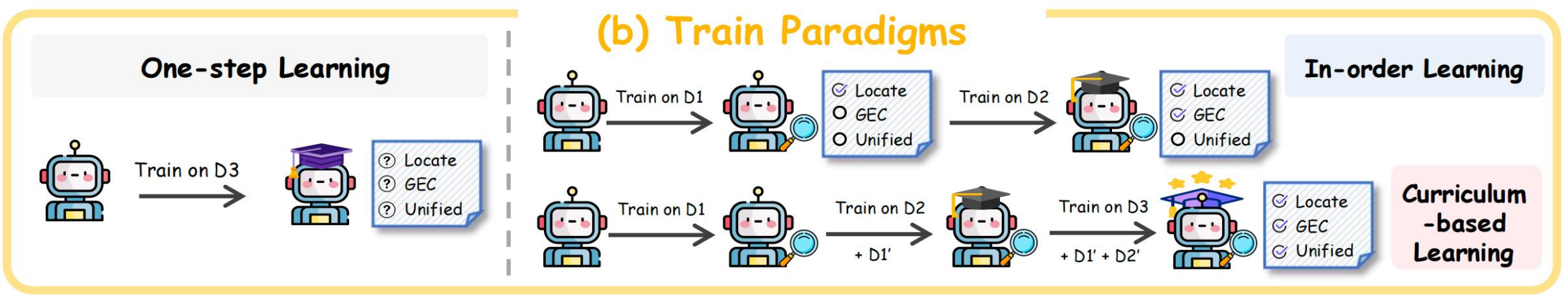
East China Normal University

Task decomposition



Character localization **GEC with oracle localization** **End-to-end edits**

Training schedules

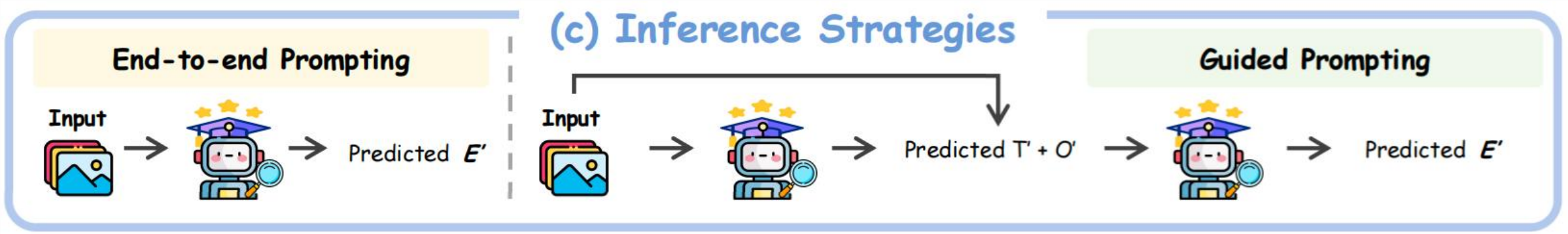


One-step Train directly on the end-to-end objective

In-order Character localization, then GEC with oracle localization

Curriculum-based End-to-end training with replay from earlier tasks

Training schedule and inference strategy



Guided prompting Predict character information, then correction edits

End-to-end prompting Generate final edits directly from the image

Curriculum learning improves strict edit F0.5

Strict Box + Text + Operation F0.5 (%)

Table 1. Performance comparison of MLLMs on VisCGEC with different inference strategies. "Guided" refers to Guided Prompting and "E2E" refers to End-to-end Prompting.

Method	Size	Inference	Box			Box + Text			Box + OP			Box + Text + OP		
			Prec.	Rec.	$F_{0.5}$	Prec.	Rec.	$F_{0.5}$	Prec.	Rec.	$F_{0.5}$	Prec.	Rec.	$F_{0.5}$
Qwen2.5-VL-7B Learned with Various Schedules														
One-step	7B	Guided	6.62	1.99	4.51	0	0	0	4.41	1.32	3.01	0	0	0
		E2E	9.82	8.39	9.50	3.62	3.09	3.50	5.94	5.08	5.75	3.62	3.09	3.50
In-order	7B	Guided	8.42	7.06	8.11	3.16	2.65	3.04	3.42	2.87	3.29	2.63	2.21	2.53
		E2E	2.67	1.99	2.50	0.59	0.44	0.56	2.08	1.55	1.94	0.59	0.44	0.56
Curriculum-based	7B	Guided	8.94	10.38	9.19	2.28	2.65	2.35	4.37	5.08	4.50	1.90	2.21	1.96
		E2E	**11.16**	**10.38**	**11.00**	**5.46**	**5.08**	**5.38**	**7.84**	**7.28**	**7.72**	**5.46**	**5.08**	**5.38**
Existing MLLMs														
GPT-4o	-	Guided	3.64	0.44	1.49	1.82	0.22	0.74	1.82	0.22	0.74	0	0	0
		E2E	0.58	0.44	0.55	0	0	0	0.29	0.22	0.27	0	0	0
Claude-sonnet-4	-	Guided	1.73	2.43	1.84	0	0	0	0.47	0.66	0.50	0	0	0
		E2E	1.53	1.77	1.57	0.19	0.22	0.20	0.76	0.88	0.79	0.19	0.22	0.20
Gemini-2.5-flash	2000B	Guided	1.05	1.10	1.06	0	0	0	0.21	0.22	0.21	0	0	0
		E2E	0.60	0.66	0.61	0	0	0	0.20	0.22	0.20	0	0	0
Doubao-1.5-pro	20B	Guided	0	0	0	0	0	0	0	0	0	0	0	0
		E2E	0.83	0.66	0.79	0.55	0.44	0.53	0.28	0.22	0.26	0.28	0.22	0.26
DeepSeek-VL2	4.5B	Guided	0	0	0	0	0	0	0	0	0	0	0	0
		E2E	0	0	0	0	0	0	0	0	0	0	0	0
Qwen2.5-VL-7B	7B	Guided	0.54	0.88	0.59	0	0	0	0.27	0.44	0.29	0	0	0
		E2E	0.38	0.88	0.42	0	0	0	0.28	0.66	0.32	0	0	0
Qwen2.5-VL-32B	32B	Guided	0.60	0.66	0.61	0.20	0.22	0.20	0.40	0.44	0.41	0	0	0
		E2E	0.47	0.22	0.38	0	0	0	0.47	0.22	0.38	0	0	0
Qwen2.5-VL-72B	72B	Guided	1.30	2.43	1.44	0.12	0.22	0.13	0.95	1.77	1.04	0.12	0.22	0.13
		E2E	0.45	0.44	0.45	0	0	0	0.23	0.22	0.22	0	0	0
Qwen-VL-max	-	Guided	5.42	3.53	4.90	1.36	0.88	1.22	3.05	1.99	2.76	1.36	0.88	1.22
		E2E	3.02	2.21	2.55	1.51	1.10	1.41	2.72	1.99	2.53	1.51	1.10	1.41

5.38

Strict edit F0.5 (%)
Curriculum + end-to-end prompting

+1.88 points over the one-step baseline

Strongest combination in this study.
Precise end-to-end correction remains difficult.

Localization remains a major bottleneck

Table 2. Performance of the methods on individual sub-tasks. "CL" denotes Character Localization, which detects every character in the image by predicting its bounding box (Box), its recognized character (Text), and whether it is a fake error (Op; *Replace* or *Keep*). "GEC with OL" denotes Grammatical Error Correction with Object Localization, and "E2E-VisCGEC" denotes End-to-End Visual GEC.

Method	Task	Box			Box + Text			Box + Op			Box + Text + Op		
		Prec.	Rec.	$F_{0.5}$	Prec.	Rec.	$F_{0.5}$	Prec.	Rec.	$F_{0.5}$	Prec.	Rec.	$F_{0.5}$
Qwen2.5-VL-7B Learned with Various Schedules													
Curriculum-based	CL	42.15	39.10	41.50	21.11	19.58	20.79	41.79	38.76	41.14	21.11	19.58	20.79
	GEC with OL	30.72	32.01	30.97	12.92	13.47	13.03	23.09	24.06	23.28	12.08	12.58	12.17
	E2E-VisCGEC	11.16	10.38	11.00	5.46	5.08	5.38	7.84	7.28	7.72	5.46	5.08	5.38
Existing MLLMs													
Qwen2.5-VL-72B	CL	18.07	16.71	17.78	2.75	2.54	2.70	17.94	16.59	17.65	2.73	2.52	2.68
	GEC with OL	28.86	25.17	28.04	14.94	13.02	14.51	19.24	16.78	18.69	14.43	12.58	14.02
	E2E-VisCGEC	0.45	0.44	0.45	0	0	0	0.23	0.22	0.22	0	0	0

Image-only correction still struggles with precise grounding

Implications for VisCGEC-v2.0

Perception and correction — Separate diagnostics help locate the source of errors

Skill integration — Train for the actual image-only test condition

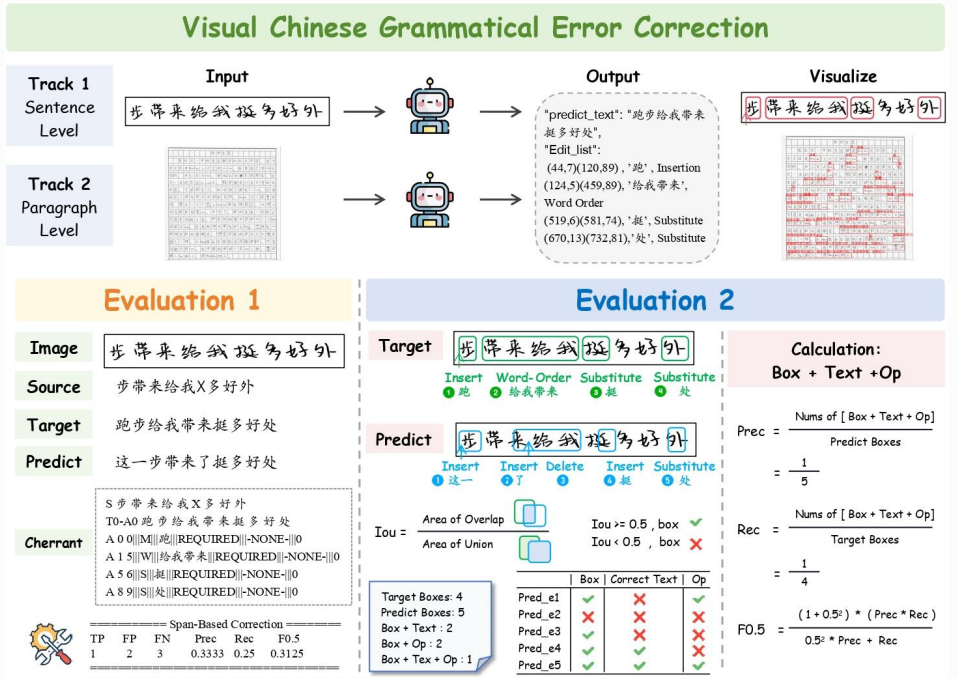
Paragraph context — Study broader context while retaining precise localization

VisCGEC-v2.0: shared challenges

Accurate, visually grounded feedback on handwritten Chinese

Sentence-Level and Paragraph-Level tracks

Corrected full text and localized edits
Joint evaluation of text and edit quality





Thank you

Next, let's hear how the top two teams designed their systems.

Tianchi: tianchi.aliyun.com/competition/entrance/532501/information